A completeness-gated method for de-risking large ERP conversions

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Doctoral dissertation defense

The thesis makes three contributions to ERP migration practice

C1 — Metric: a field-level completeness measure over the data dictionary.

C2 — Evidence: completeness predicts SUM defect rate across 126 tables.

C3 — Method: a completeness gate that blocks premature cutover.

The central question links a measurable cause to a costly effect

Can a field-level completeness metric, used as a cutover gate,
reduce ERP conversion defect rates at acceptable cost?

A design-science study builds the metric and tests it on a live migration

Methods

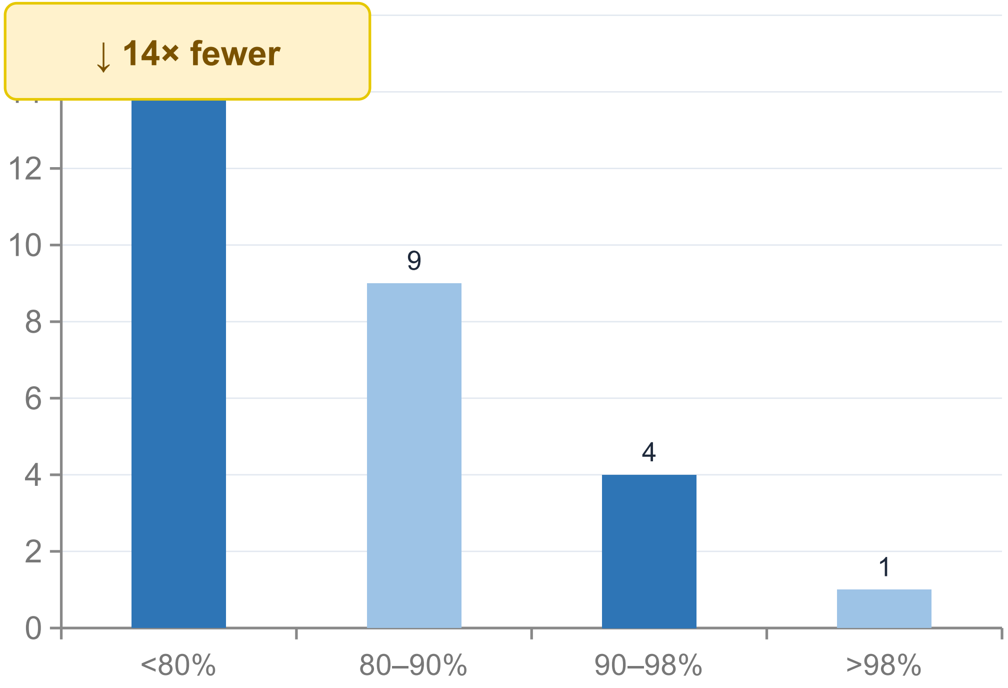
Build: completeness metric over PM + PP-PI dictionaries.

Evaluate: correlate completeness with logged SUM defects.

Validate: expert review with the migration office.

Identification + threats to validity → Appendix B

Defect rate falls 14× across completeness bands, supporting the gate



What to take away

- Strong monotonic relationship

Gate at ≥98% removes nearly all defects

Consistent across both modules

Robust to band-width choice (Appendix B)

SUM run logs, Project NEO (2026)

The gate is causal-plausible and cheap, but tested on a single programme

Interpretation

Mechanism: unmapped fields are the proximate cause of run errors.

Cost: dictionary completion << cost of a failed cutover.

Limitation

Single case: replication across clients is future work.

Conclusions

1. **A measurable gate works:** completeness predicts defects.

2. **The effect is large:** 14× across completeness bands.

3. **Practice implication:** gate cutover on field completeness.

References

Hevner, A.R. et al. (2004). Design Science in Information Systems Research. MIS Quarterly, 28(1).

SAP SE (2025). Software Update Manager (SUM) Guide. SAP Help Portal.

Halif, S. (2026). Project NEO Migration Cockpit. CBC Israel.

Defect–completeness relationship is stable across band definitions

Bandwidth: tertile and quartile splits give the same ordering.

Module split: holds separately for PM and PP-PI.